CSCE 763: Digital Image Processing

Spring 2026

Dr. Yan Tong

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University of South Carolina

Course Information

Instructor: Dr. Yan Tong

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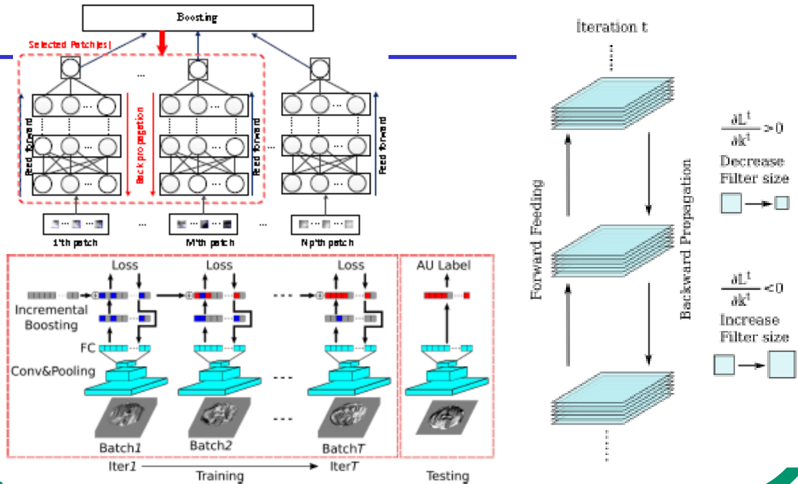
Office: Storey Innovation Center 2273

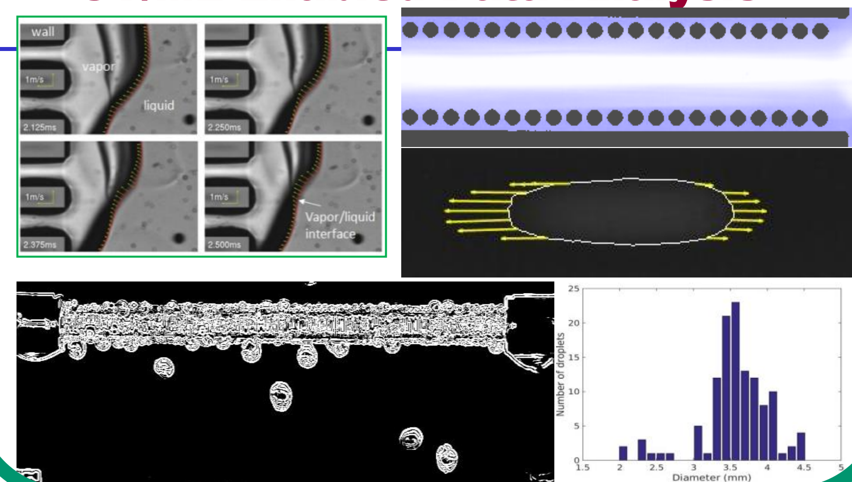
Office Hours: 10am -11am, Tuesday

Dr. Tong's Main Research Areas

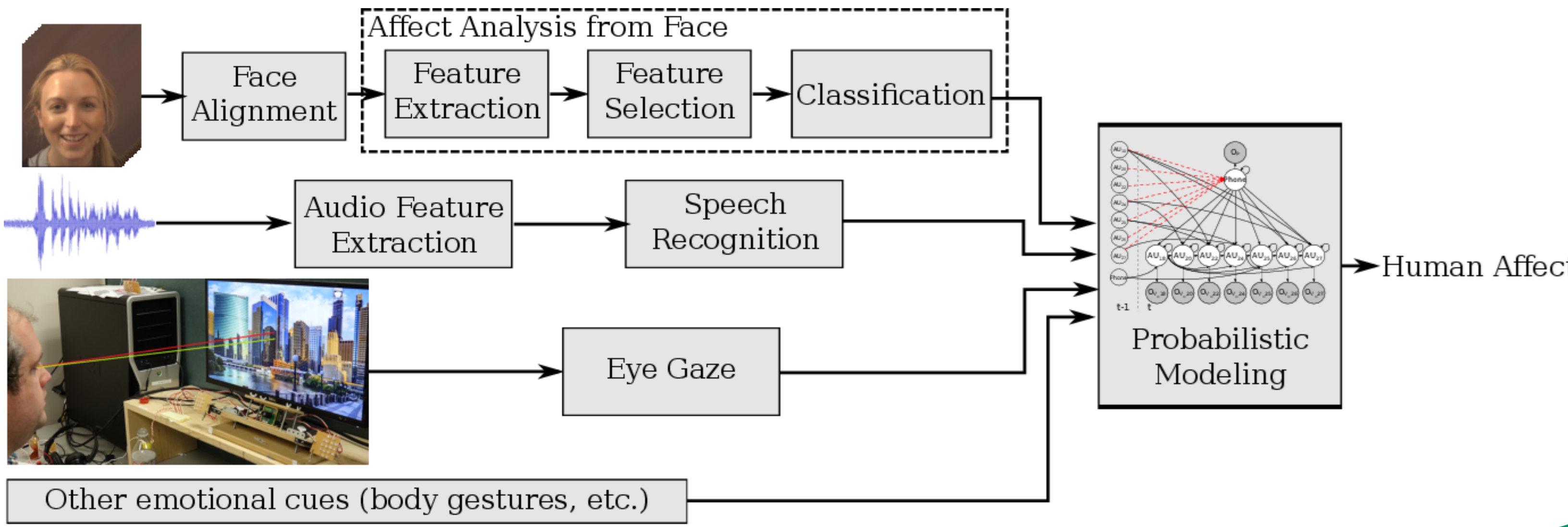
Fundamental Research in CV/ML



CV/ML Enabled Data Analysis



Multimodal Information Fusion



Now, tell me about yourself!

- Name
- Major
- Research interest
- Why do you take this course

Today's Agenda

- Welcome
- Tentative Syllabus
- Topics covered in the course

Class Communication

Class website

http://www.cse.sc.edu/~tongy/csce763/csce763.html

Blackboard Ultra

Tentative Syllabus

- Prerequisites
- Objectives
- Textbook
- Grade

Prerequisites of This Course

This is a computer science course

- It will involve a fair amount of math
  - calculus, linear algebra, geometry
  - probability
  - analog/digital signal processing
  - graph theory etc.

- It will involve the modeling and design of a real system - one final course project
  - Programming skills with MATLAB, Python, or C++

The Objective of This Course

This is a graduate-level topic course

- Research oriented
  - Paper reading & presentation
  - Final project & presentation
  - Review on the state-of-the-art
- Understanding → Innovation
  - your own innovative and original work/opinion/result
- Basic knowledge → Research frontier
  - learn through reading recent papers

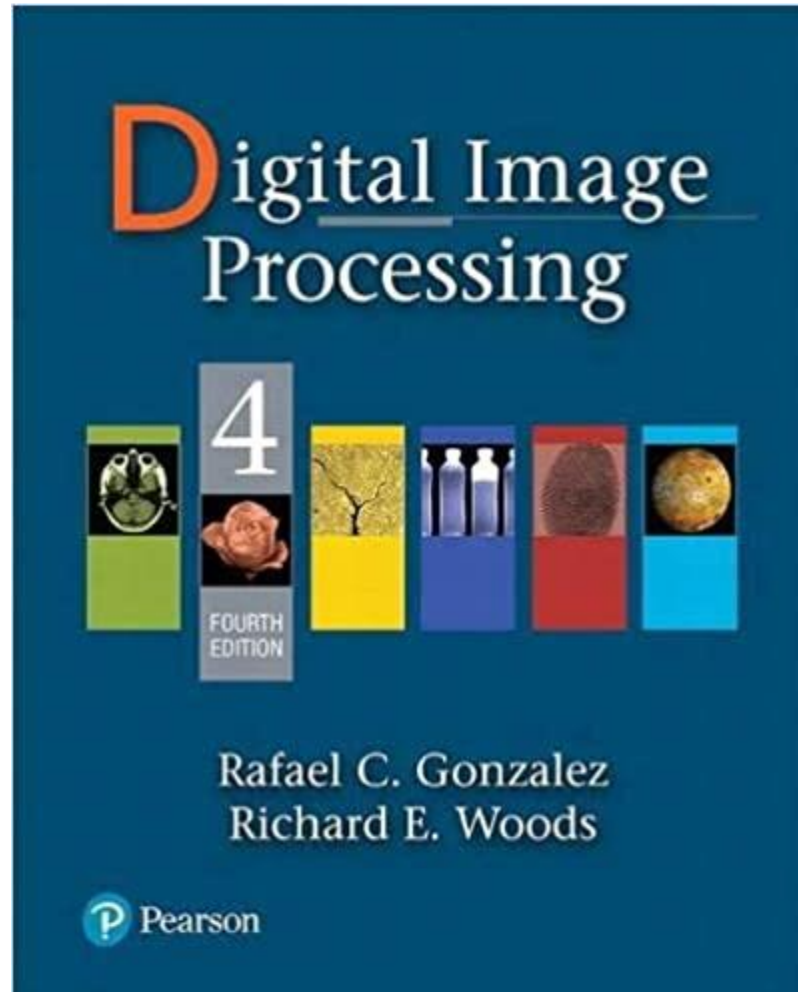
Textbook

Required:

Digital Image Processing, **Rafael C. Gonzalez and Richard E. Woods, 4th Edition, Pearson**

We will cover many topics in this textbook

We will also include special topics on recent progresses on image processing



Others

Department seminars

Guest lectures

Requirement for Final Project

Option 1: A complete research project

- Introduction (problem formulation/definition)
- literature review
- the proposed method and analysis
- experiment
- conclusion
- reference

Option 2: A survey research

- A well-defined problem or topic
- a complete list of previous (typical) work on this problem (15+ papers under the topic)
- clearly and briefly describe the topic
- analyze each method/group and compare them
- give the conclusion and list of references

Requirement for Final Project

Requirements

- Select a topic and write a one-page proposal (due Sunday, Feb 22)
- Progress report (discuss with the instructor)
- Research work and report writing
- Oral presentation
- Final project report

Requirement for Final Project

Teamwork is acceptable for a research project (Option 1)

- <=2 people
- Get the permission from the instructor first
- Under a single topic, each member must have their own specific tasks
- One combined report with each member clearly stating their own contributions
- One combined presentation

Requirement for Final Project

Written report

- Report format: the same as an IEEE conference paper
- Executable code must be submitted with clear comments except for a survey study

Academic integrity (avoiding plagiarism)

- don’t copy other person’s work
- describe using your own words
- complete citation and acknowledgement whenever you use any other work (either published or online)

Requirement for Final Project

Evaluation

- written report (be clear, complete, correct, etc.)
- code (be clear, complete, correct, well documented, etc.)
- oral presentation
- discussion with the instructor
- quality: publication-level project – extra credits

Requirement for Final Project

Notes:

- You are encouraged to incorporate your own research expertise in, but the project topic must be related to the content of this course
- Discuss with the instructor on topic selection, progress, writing, and presentation
- Use the library and online resource

Paper Reading and Presentation

- A paper picked by yourself and approved by the instructor
  - Suggested paper source: To be provided
- Thorough understanding of the paper
- Prepare PPT slides
  - Clearly explain the main contributions in the selected paper
  - Critical comments and discussions
- About 10 mins oral presentation for each student

Generative Artificial Intelligence Policy

In this course, students shall give credit to AI tools whenever used, even if only to generate ideas rather than usable text or illustrations. When using AI tools, add an appendix showing

- the entire exchange, highlighting the most relevant sections;
- a description of precisely which AI tools were used (e.g. ChatGPT private subscription version or DALL-E free version),
- an explanation of how the AI tools were used (e.g. to generate ideas, turns of phrase, elements of text, long stretches of text, lines of argument, pieces of evidence, maps of the conceptual territory, illustrations of key concepts, etc.);
- an account of why AI tools were used (e.g. to save time, to surmount writer’s block, to stimulate thinking, to handle mounting stress, to clarify prose, to translate text, to experiment for fun, etc.).

Use AI tools during quizzes, exams, or homework assignments is NOT allowed unless explicitly permitted and instructed.

Major Topics Covered in Class

Image acquisition and digital image representation

Image enhancement

Image restoration

Color image processing

Image compression

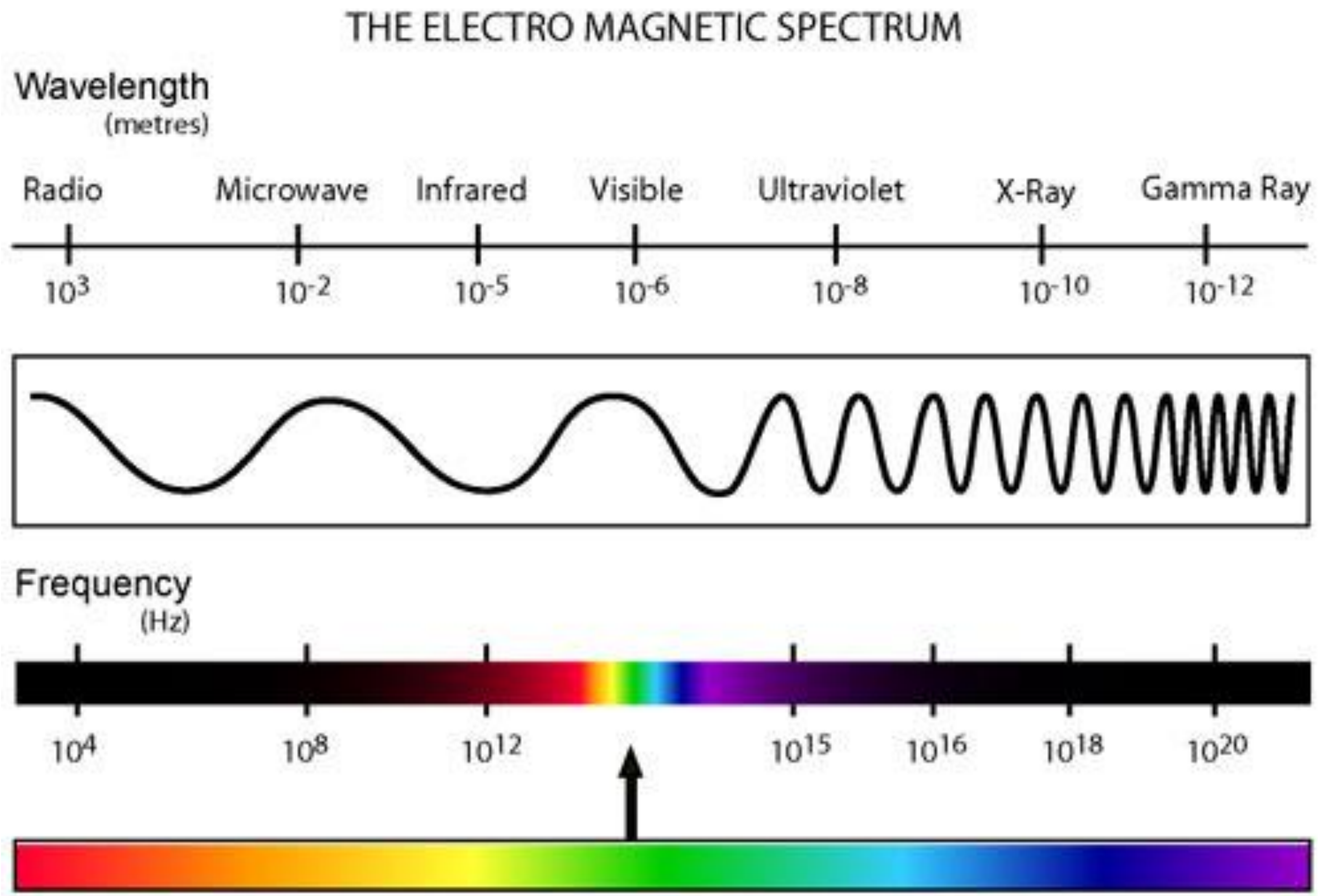
Image segmentation

Morphological image processing

Special topics on recent progresses on digital image processing

Human Perception VS Machine Vision

- **Limited vs entire EM spectrum**



http://www.kollewin.com/blog/electromagnetic-spectrum/

Image Processing → Image Analysis

Low level	Image acquisition
	Image enhancement
	Image compression
Mid level	Image segmentation
	Object recognition
High level	Scene understanding
	Semantics

Image processing

Image analysis
(Computer vision,
Pattern recognition, etc.)

Image Acquisition and Representation

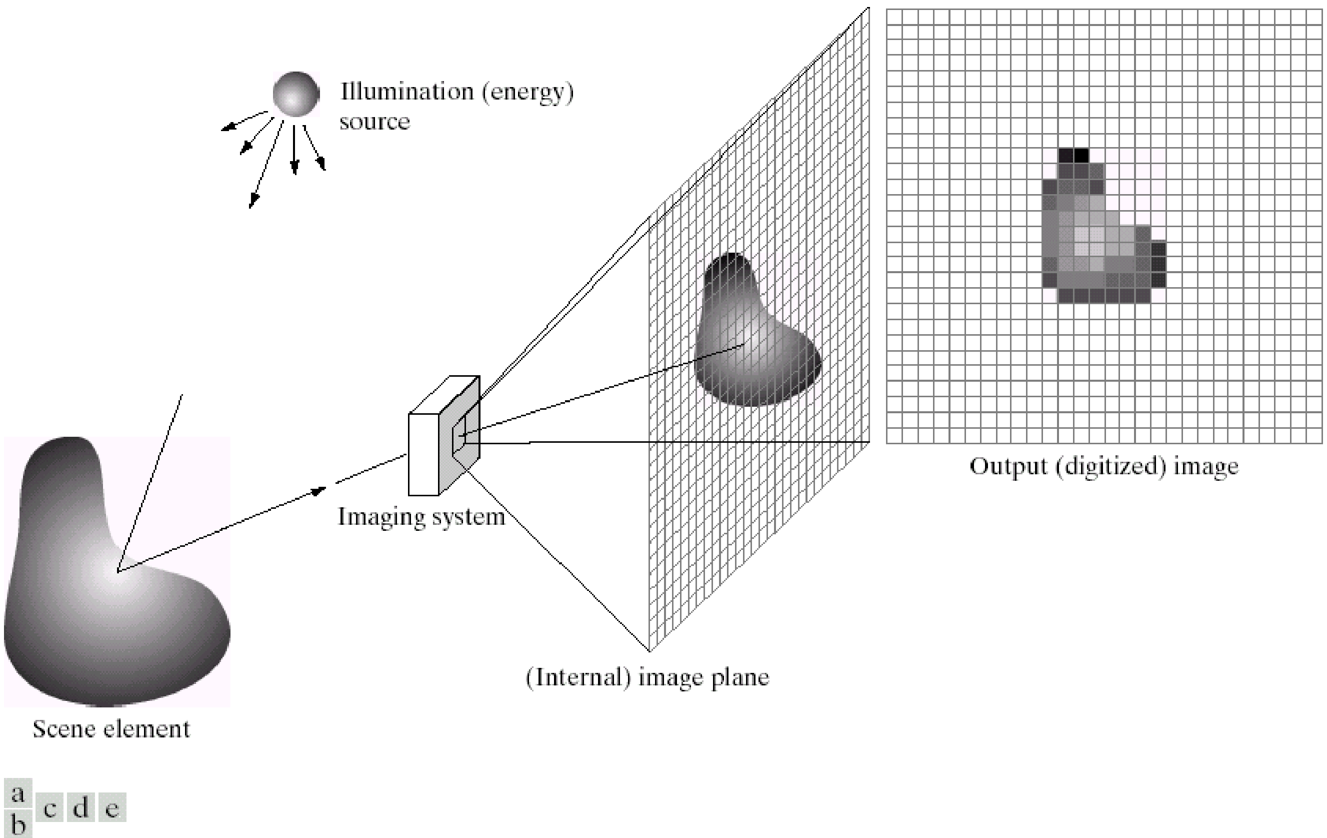
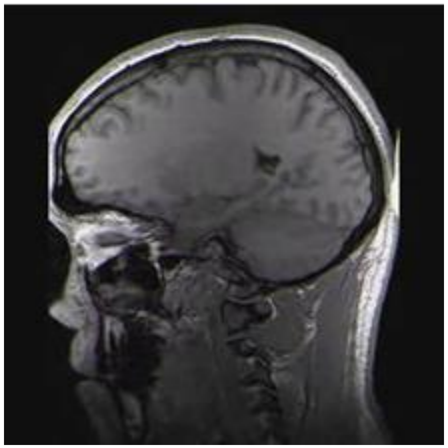
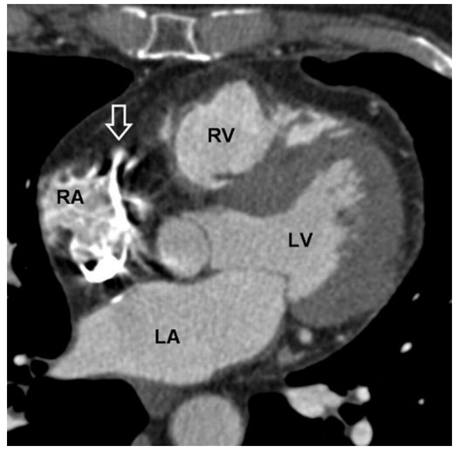


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Examples

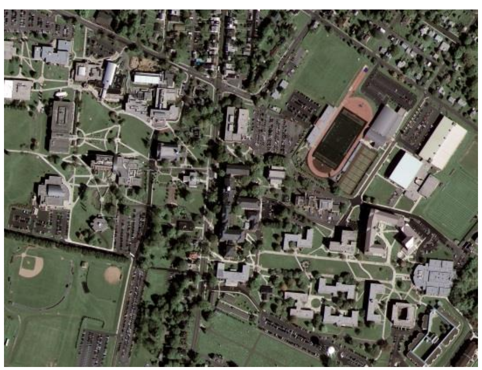


1. Brain MRI

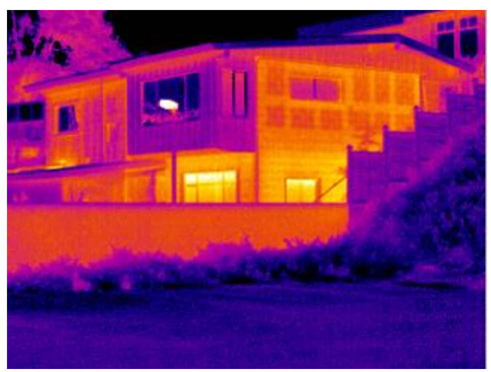


2. Cardiac CT

3. Fetus Ultrasound



4. Satellite image



5. IR image

1 and 3. http://en.wikipedia.org
2. http://radiology.rsna.org
4. http://emap-int.com
5. http://www.imaging1.com

Image Acquisition

Camera + Scanner → Digital Camera: Get images into computer

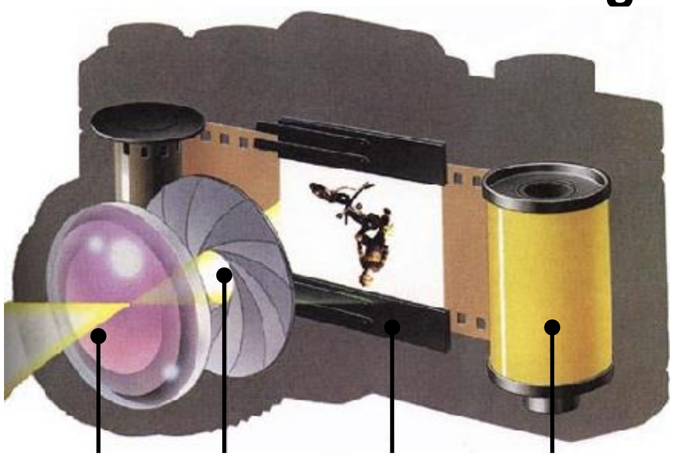
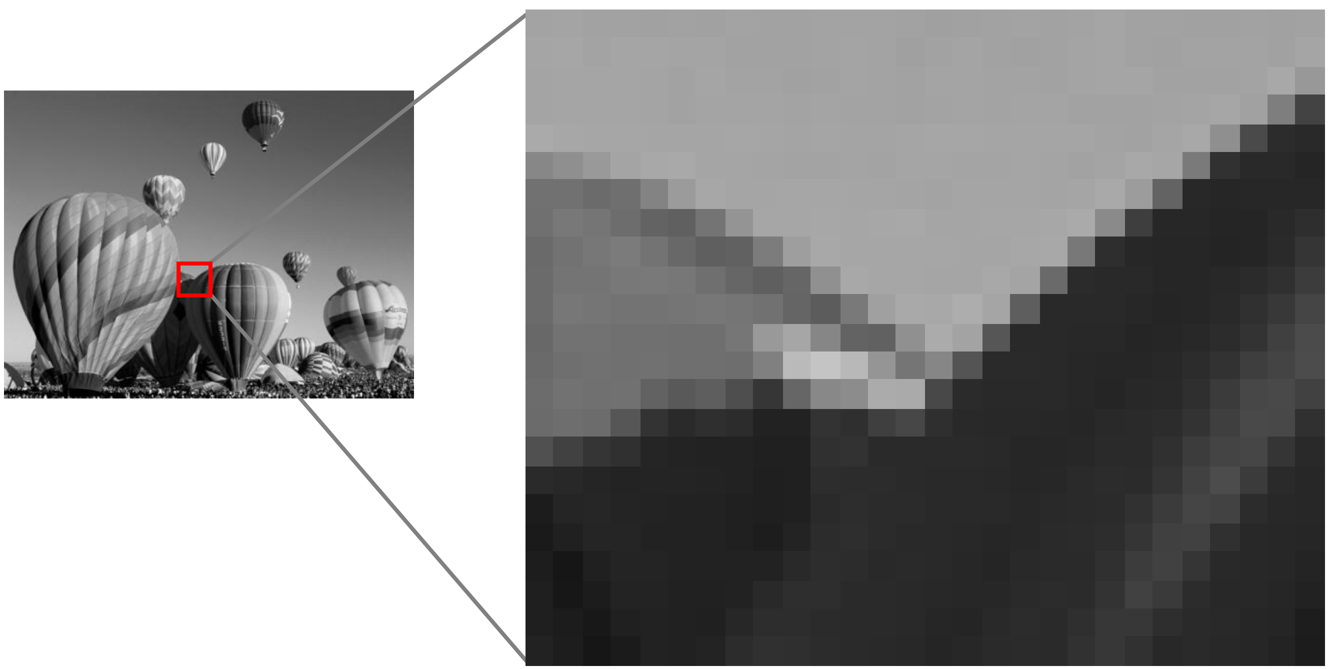
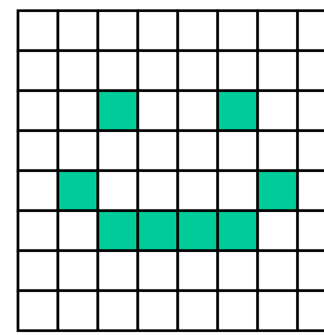


Image Representation

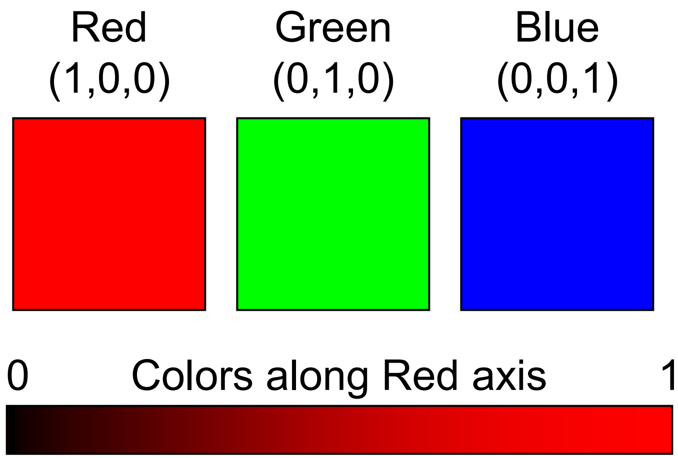
Discrete representation of images

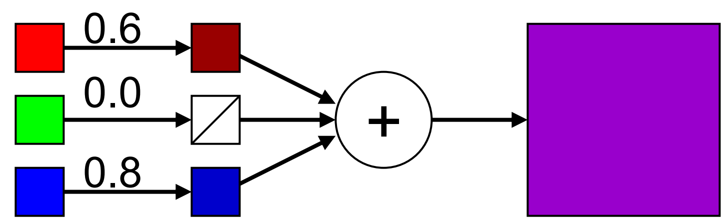
- we'll carve up image into a rectangular grid of pixels $P[x,y]$
- each pixel p will store an intensity value in [0 1]
- 0 → black; 1 → white; in-between → gray
- Image size mxn → (mn) pixels





Color Image


Red
(1,0,0)
Green
(0,1,0)
Blue
(0,0,1)
0
Colors along Red axis
1


0.6
0.0
0.8
+


RGB
channels

Video: Frame by Frame

30 frames/second

Image Enhancement

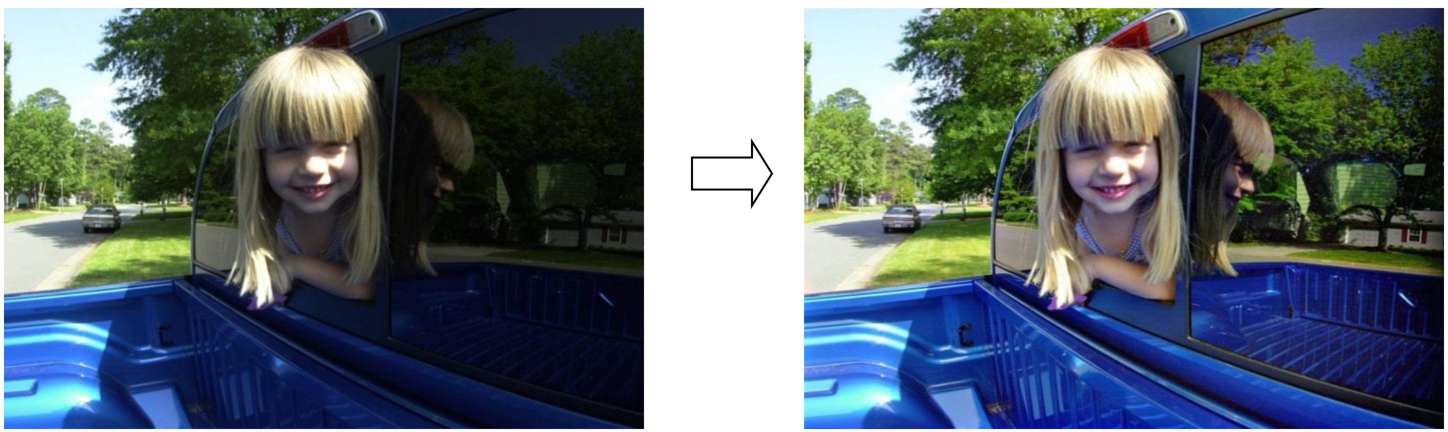
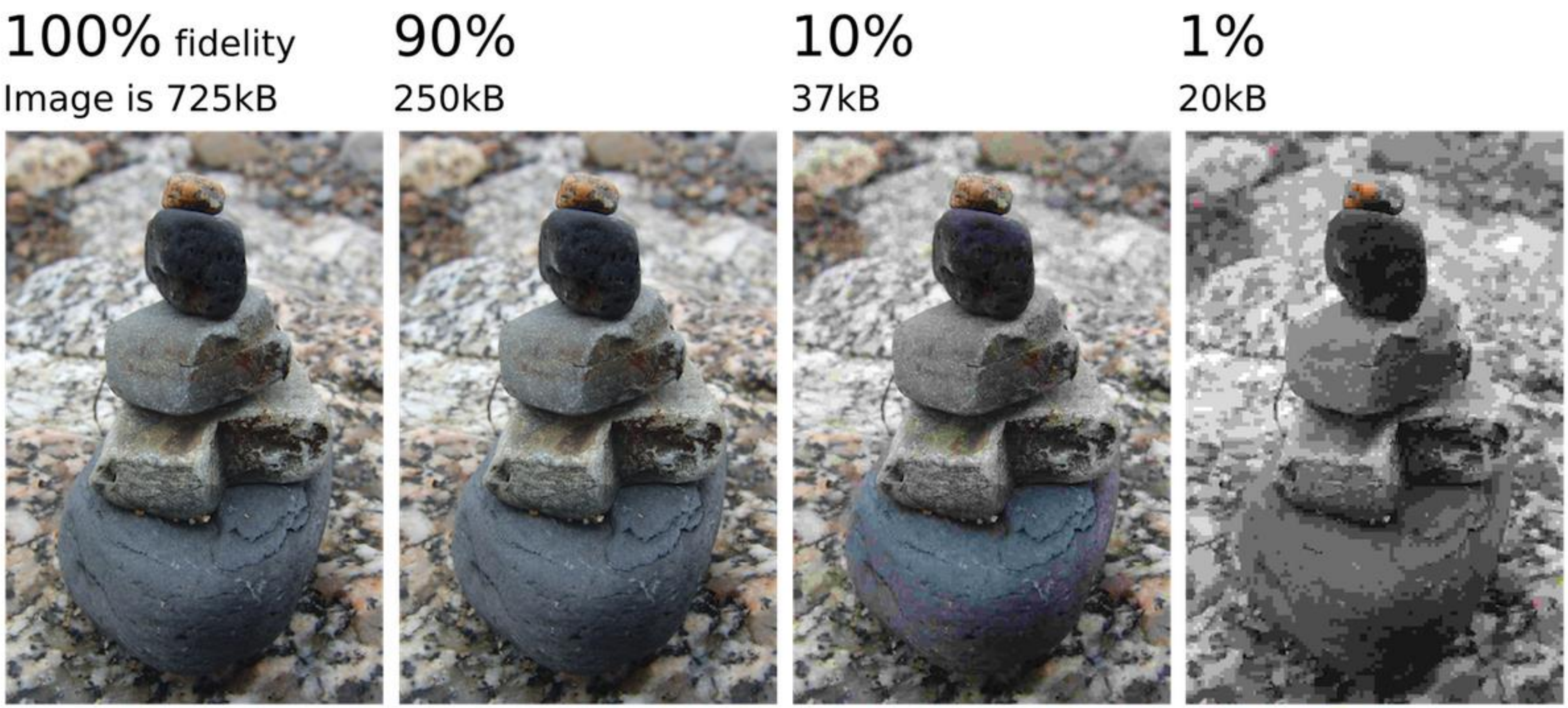


Image Restoration

Image Compression



→ **Video compression**

Image Processing → Image Analysis

Low level

Mid level

High level

- Image acquisition
- Image enhancement
- Image compression
- Image segmentation
- Object recognition
- Scene understanding
- Semantics

Image processing

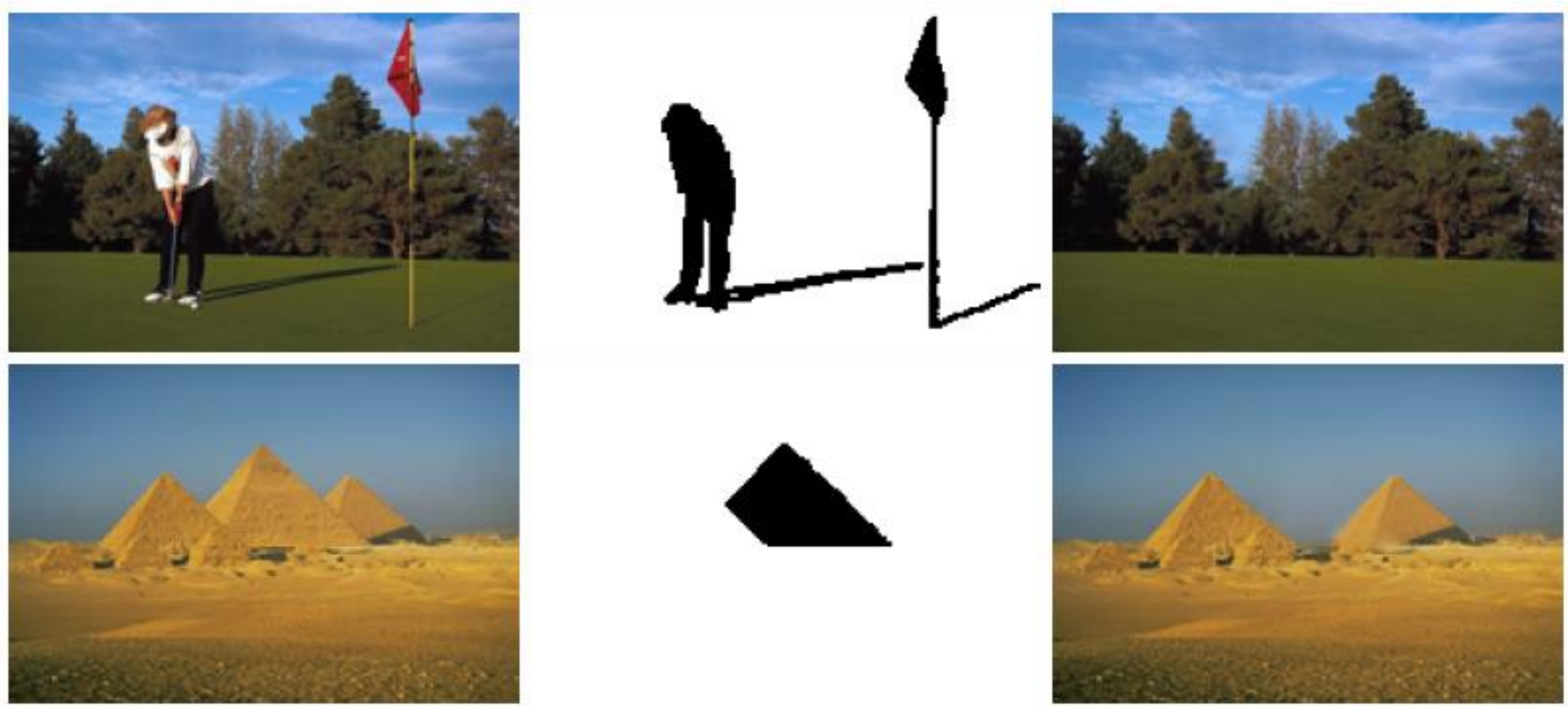
Image analysis
(Computer vision,
Pattern recognition, etc.)

Image Segmentation

Microsoft multiclass segmentation data set

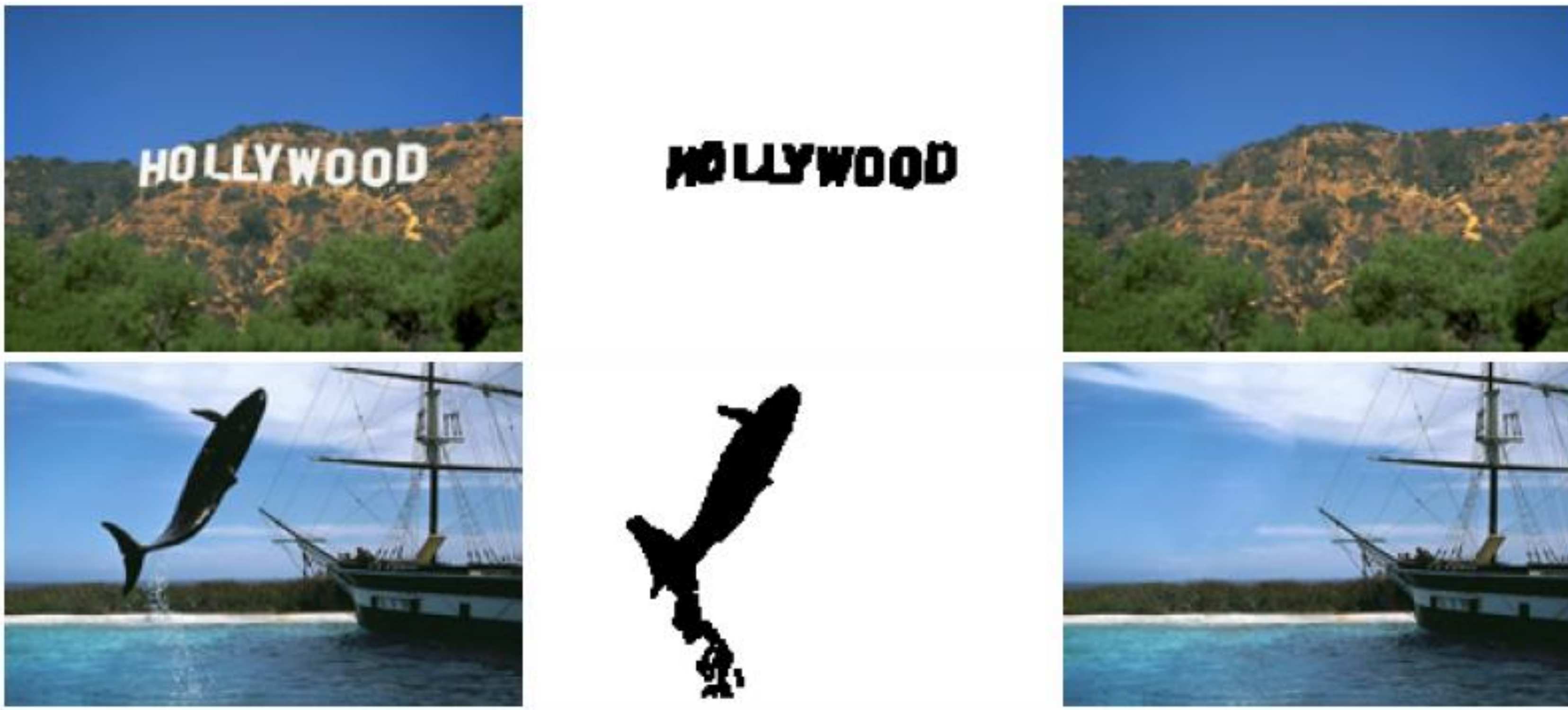
Image Completion

Interactively select objects. Remove them and automatically fill with similar background (from the same image)

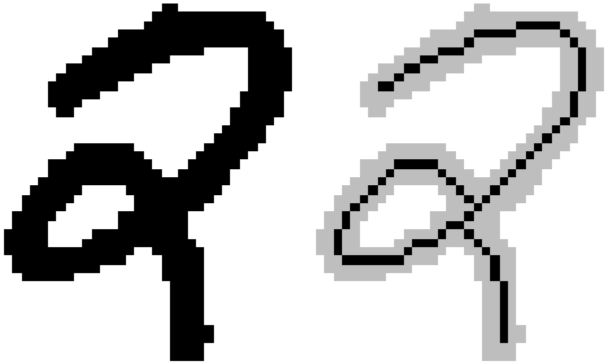


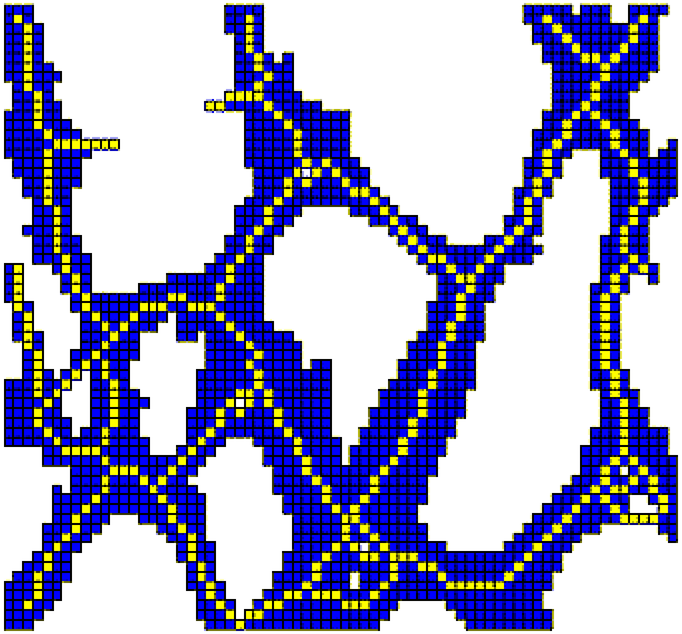
I. Drori, D. Cohen-Or, H. Yeshurun, SIGGRPAH'03

More Examples

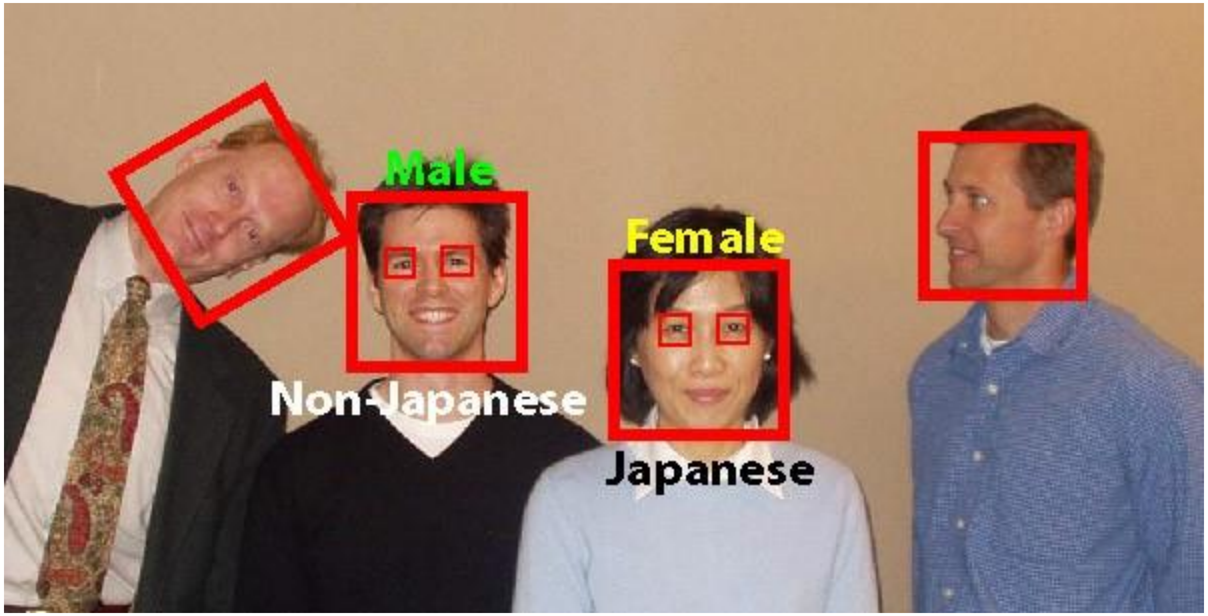


Morphological Image Processing

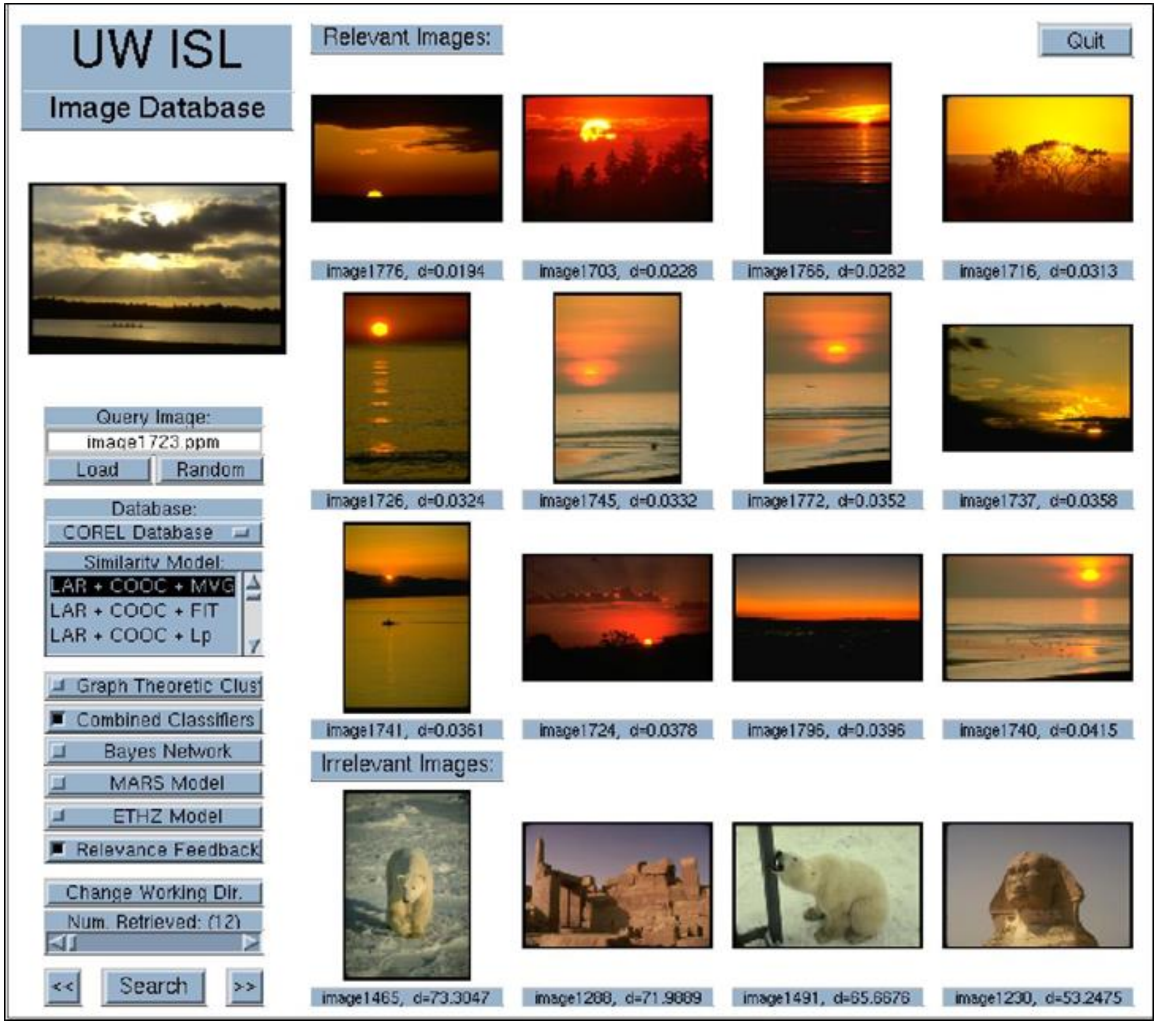




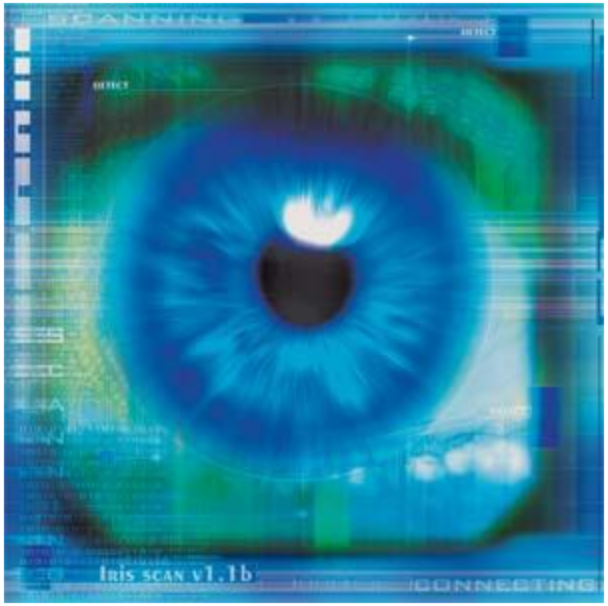
Object Detection / Recognition

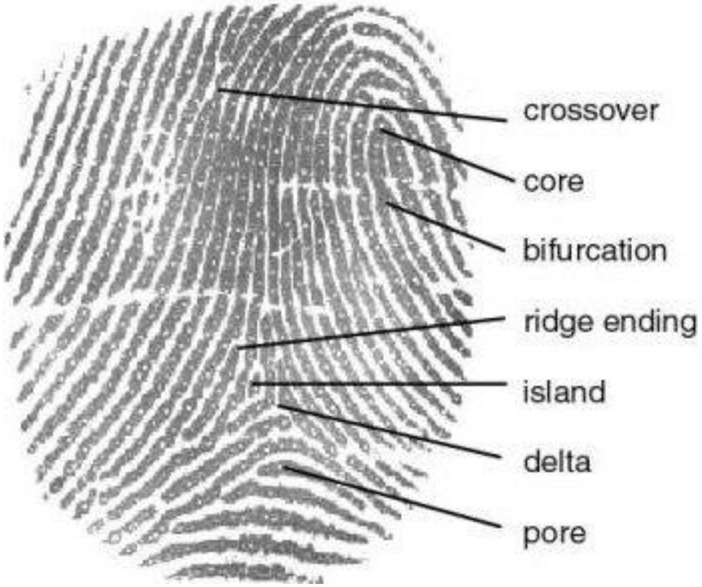


Content-Based Image Retrieval

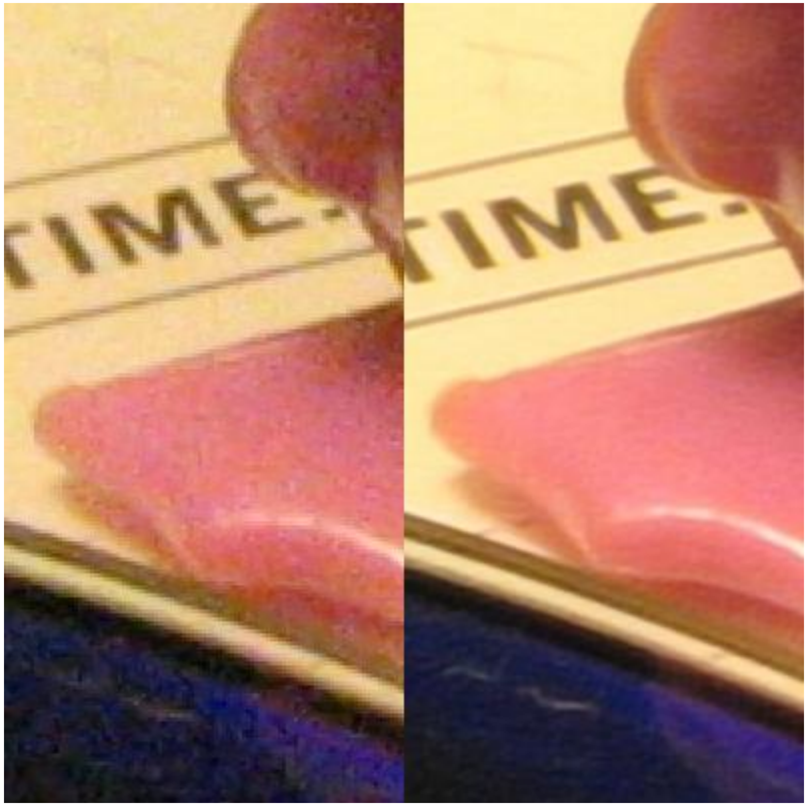


Biometrics


Iris scan v1.1b


crossover
core
bifurcation
ridge ending
island
delta
pore

Super-Resolution



Applications of Digital Image Processing

Digital camera

Photoshop

Human computer interaction

Medical imaging for diagnosis and treatment

Surveillance

Automatic driving

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Fast-growing market!